

Supplemental Information

Cell-Type-Specific Interleukin 1

Receptor 1 Signaling in the Brain

Regulates Distinct Neuroimmune Activities

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Figure S1. Examination of IL-1R1 mRNA and Cre expression in cell-type specific Cre and IL-1R1 lines. Related to Figure 1 and Figure 2.

(A-B) Results of ISHH labeling of brain sections from WT (A) and *Il1r1^{tr}* mice (B). IL-1R1 riboprobe labeling was found in the endothelial cells (A, left), choroid plexus cells (A, middle) and ependymal cells (A, right) in the WT mice and was absent in the dentate gyrus (B, left), choroid plexus cells (B, middle) and endothelial cells (B, right) in the *Il1r1^{tr}* mice. Scale bar: 100 μ m. Dashed square marks the area shown at higher magnification on the bottom right.

(C) IHC labeling of DAPI, RFP and CD31 of hippocampal sections from WT mice. Scale bar: 100 μ m. RFP is not present in these mice without the knockin tdTomato.

(D-F) IHC labeling of the reporter for *Tie2*-Cre (D, middle), *Cx3cr1*-Cre (E, middle), and *Gfap*-Cre (F, middle) with cell-type specific markers Ly6C (D, left), Iba1 (E, left) and GFAP (F, left) of brain sections from *Tie2*-Cre-R26tdT, *Cx3cr1*-Cre-R26tdT and *Gfap*-Cre-R26tdT mice. Scale bar for (D-E): 100 μ m. Scale bar for (F): 50 μ m.

(G-H) Representative tilescan images of tdTomato (G) and eGFP (H) labeling of hippocampus sections from *Camk2a*-Cre-R26tdT and *Vglut2*-Cre-R26tdT mice. Scale bar: 200 μ m.

(I) IHC labeling of NeuN and RFP of brain sections from *Gfap*-Cre-*Il1r1^{tr}* mice. Scale bar: 100 μ m.

(J) Representative tilescan images of RFP labeling of *Vglut2*-Cre-*Il1r1^{tr}* hippocampal sections. Scale bar: 200 μ m.

(K-L) Representative images of RFP labeling of brain sections from *Gfap*-Cre-*Il1r1^{tr}* mice after sham/IL-1 injections. Scale bar: 50 μ m.

(M) IHC labeling of Iba1 and RFP of hippocampal sections from *Cx3cr1*-Cre-*Il1r1^{tr}* mice. Scale bar: 100 μ m.

(N) IHC labeling of RFP, PDGFR β and CD31 of brain sections from *Tie2*-Cre-*Il1r1^{tr}* mice. Arrow indicates a PDGFR β -labeled pericyte cell body. Dashed square marks the area shown for RFP and CD31 co-localization (top right) or RFP and PDGFR β (bottom right) double labeled images at higher magnification. Scale bar: 25 μ m.

(O) qPCR analysis of inflammatory genes in hippocampus samples of WT and *Tie2*-Cre-*Il1r1^{tr}*COX-2^{tr} mice following ICV sham/IL-1 β injections. n = 4 mice/genotype, [3 independent experiments](#).

Error bars represent the mean $\pm$ SEM. Means with asterisk (*) are significantly different from the corresponding control group ($p < 0.05$) according to *F*-protected post hoc analysis.

Figure S2. eIL-1R1 and nIL-1R1 cause distinct neuroinflammatory responses. Related to Figure 3.

(A) Representative images of CD45 and Iba1 labeling on injection side of DG sections from WT, *Il1r1^{tr}*, *Tie2*-Cre-*Il1r1^{tr}* and *Vglut2*-Cre-*Il1r1^{tr}* mice 6 days after high dose ad-ssIL-1 injections. Dashed square marks the area shown at higher magnification on the bottom right. Scale bar: 100 μ m.

(B-C) Quantification of Iba1 proportional area (B) and CD45⁺ cells (C) on injection side and contralateral side of DG sections from WT, *Il1r1^{tr}*, *Tie2*-Cre-*Il1r1^{tr}* and *Vglut2*-Cre-*Il1r1^{tr}* mice 6 days after ad-ssIL-1 injections. n = 4 mice/genotype, [2 independent experiments](#).

(D) Representative images of CD45 and Iba1 labeling on injection side of DG sections from WT, *Il1r1^{tr}*, *Tie2*-Cre-*Il1r1^{tr}* and *Vglut2*-Cre-*Il1r1^{tr}* mice 6 days after low dose ad-ssIL-1 injections. Dashed square marks the area shown at higher magnification on the bottom right. Scale bar: 100 μ m.

(E) Quantification of Iba1 proportional area on injection side and contralateral side of DG sections from WT, *Il1r1^{tr}*, *Tie2*-Cre-*Il1r1^{tr}* and *Vglut2*-Cre-*Il1r1^{tr}* mice 6 days after ad-ssIL-1 injections. n = 4 mice/genotype, [2 independent experiments](#).

Error bars represent the mean $\pm$ SEM. Means with asterisk (*) are significantly different from the corresponding control group ($p < 0.05$) according to *F*-protected post hoc analysis.

Figure S3. Microglial activation and leukocyte recruitment occur independently in response to IL-1. Related to Figures 4-6.

(A-E) Influence of leukocyte depletion by vinblastine/cyclophosphamide on central IL-1 induced microglial morphological alterations. (F-H) Influence of microglia depletion by PLX on central IL-1 induced leukocyte recruitment.

(A) Representative images of CD45 and Iba1 labeling of hippocampus sections from saline, vinblastine or cyclophosphamide treated WT mice 24 h after ICV IL-1 β injections. Scale bar: 100 μ m.

(B-D) Quantification of Iba1 proportional area (B), average microglial total process length (C), and microglial soma size (D) in DG sections from saline, vinblastine or cyclophosphamide treated WT mice 24 h after ICV IL-1 β injections. $n = 4-5$ mice/genotype, [3 independent experiments](#).

(E) Quantification of parenchymal CD45⁺ cell number in the DG sections from saline, vinblastine or cyclophosphamide treated WT mice 24 h after ICV IL-1 β injections. $n = 4$ mice/group, [3 independent experiments](#).

(F) Quantification of parenchymal Iba1⁺ cell number in brain sections from PLX treated WT mice after ICV IL-1 β injections. $n = 4$ mice/group, [2 independent experiments](#).

(G) Cell counts of microglia and infiltrating leukocytes in the brain by FACS analyses. $n = 3$ mice/group, [2 independent experiments](#).

(H) Representative images of CD45 and Iba1 labeling of brain sections from PLX treated WT mice 24 h after ICV sham/IL-1 β injection. Scale bar: 100 μ m.

Error bars represent the mean $\pm$ SEM. Means with asterisk (*) are significantly different from the corresponding control group ($p < 0.05$) according to *F*-protected post hoc analysis. Vinbl, vinblastine; Cyclo, cyclophosphamide.

Figure S4. Influence of depletion of circulating leukocytes on microglial morphology and influence of depletion of brain microglia on leukocyte trafficking. Related to Figures 4-6.

(A) Representative flow bivariate dot plots of CD11b, Ly6C and Ly6G labeling on enriched blood monocytes and neutrophils from saline, vinblastine or cyclophosphamide treated WT mice.

(B) Cell counts of total leukocytes, CD11b⁺Ly6G⁺Ly6C^{int} neutrophils and CD11b⁺Ly6G⁺Ly6C^{hi} monocytes in the blood from saline, vinblastine or cyclophosphamide treated WT mice. $n = 4$ mice/group, [4 independent experiments](#).

(C) Representative images of CD45 and Iba1 labeling of the hippocampus sections from saline, vinblastine or cyclophosphamide treated mice which received sham injections. Scale bar: 100 μ m.

(D-E) Quantification of Iba1 proportional area of medial orbital cortex (D) and septal area (E) sections from saline, vinblastine or cyclophosphamide treated WT mice which received sham/IL-1 β injections. $n = 4$ mice/group, [3 independent experiments](#).

(F) Cell counts of whole brain monocytes, lymphocytes and neutrophils by flow cytometry in PLX or control diet treated WT mice after saline or LPS injections. $n = 6-8$ mice/group. SAL, saline, [2 independent experiments](#).

Error bars represent the mean $\pm$ SEM. Means with asterisk (*) are significantly different from the corresponding control group ($p < 0.05$) according to *F*-protected post hoc analysis.

Figure S5. Mechanistic study of central IL-1 induced leukocyte infiltration. Related to Figure 4.

(A) Schematic of the generation of *Il1r1*^{tr/tr} bone marrow chimera (BM^{*Il1r1*^{tr/tr} $\rightarrow$ *Il1r1*^{tr/tr}} and BM^{*Il1r1*^{tr/tr} $\rightarrow$ *Tie2*^{*Cre-Il1r1*^{tr/tr}}} mice.

(B) IHC labeling of Iba1 and CD45 of brain sections from BM^{Il1r1^{fl/r}→Il1r1^{fl/r}} and BM^{Il1r1^{fl/r}→Tie2-Cre-Il1r1^{fl/r}} mice 24 h after ICV IL-1 β injections. Dashed square marks the area shown at higher magnification on the top left. Scale bar: 50 μ m.

(C) Ratio of infiltrating CD11b⁺CD45⁺ leukocytes relative to numbers of microglia in whole brains from BM^{Il1r1^{fl/r}→Il1r1^{fl/r}} and BM^{Il1r1^{fl/r}→Tie2-Cre-Il1r1^{fl/r}} mice. n = 4 mice/genotype, [2 independent experiments](#).

(D) IHC labeling of CD45 and CD31 of hippocampal sections from ICV IL-1 β injected WT, Tie2-Cre-Il1r1^{fl/r} and Tie2-Cre-Il1r1^{fl/fl} mice. Scale bar: 100 μ m.

(E) IHC labeling of RFP, GFAP, Iba1, CD31 and CD206 of brain sections from CCL2RFP^{flox} mice after ICV sham/IL-1 β injections. Dashed square marks the area shown at higher magnification on the bottom right. An orthogonal view of each higher magnification, double-labeled, z-stack image is presented on the far right. White lines represent vertical or horizontal optical cuts through the stack. Scale bar for top three panels: 100 μ m. Scale bar for bottom three panels: 50 μ m.

(F) Relative CCL2 mRNA levels of brain tissues from WT mice after PLX treatment and the rest mouse lines 3 h following ICV IL-1 β injections, complementary to IL-1R1 lines shown in Figure 5F. n = 4-5 mice/genotype, [3 independent experiments](#).

Error bars represent the mean $\pm$ SEM. Means with different letters (a, b, c, d) show groups that are significantly different ($p < 0.05$) according to *F*-protected post hoc analysis.

Figure S6. Central IL-1 induces CAM expression in IL-1R1⁺/IL-1R1⁻ endothelial and ependymal cells. Related to Figure 5.

(A-B) IHC labeling of RFP (A-B), ICAM-1 (A) and VCAM-1 (B) of brain sections from Il1r1^{GR/GR} mice after ICV IL-1 β injections. Scale bar: 100 μ m.

(C) Quantification of RFP and ICAM/VCAM colocalization of brain sections from Il1r1^{GR/GR} mice after ICV IL-1 β injections. n = 3 mice/group, [2 independent experiments](#).

(D-E) Representative tilescan images of VCAM-1 (D) and ICAM-1 (E) labeling of brain sections from WT, Tie2-Cre-Il1r1^{fl/r} and Tie2-Cre-Il1r1^{fl/fl} mice after sham/IL-1 β injections. Arrows point to ICAM expression in ependymal cells at the lateral ventricle. Scale bar: 200 μ m.

Figure S7. Cell-type specific IL-1R1s mediate changes in microglial morphology and gene expressions. Related to Figure 6.

(A-B) Quantification of average microglial total process length (A) and Iba1⁺ cell density (B) of DG sections from cell-type specific IL-1R1 lines 24 h after ICV IL-1 β injections. n = 4-5 mice/genotype, [5 independent experiments](#).

(C) qPCR analysis of NOS2 in hippocampus samples of cell-type specific IL-1R1 lines and PLX treated WT mice 3 h following ICV sham/IL-1 β injections. n = 4-5 mice/genotype, [6 independent experiments](#).

(D) qPCR analysis of inflammatory genes in of Il1r1^{fl/fl}, Tie2-Cre-Il1r1^{fl/fl} and Gfap-Cre-Il1r1^{fl/fl} mice 3 h following ICV sham/IL-1 β injections. n = 4 mice/genotype, [4 independent experiments](#).

(E) IHC labeling of RFP and GFAP of striatum sections from ICV IL-1 β injected AAVCre-Il1r1^{fl/r} mice. Dashed square marks the area shown at higher magnification on the bottom. Scale bar: 100 μ m.

(F) IL-6 mRNA levels of sham injected, control side (AAVGFP) and aIL-1R1 KO side (AAVCre) striatum in AAVCre-Il1r1^{fl/fl} mice. n = 4-5 mice/genotype, [1 independent experiment](#).

(G-H) Quantification of mean fluorescence intensity (MFI) of CD86 (G) and CD11b (H) in microglia from ICV IL-1 β injected WT, Tie2-Cre-Il1r1^{fl/r} and Tie2-Cre-Il1r1^{fl/fl} mice by flow cytometric analysis. n = 4 mice/per genotype, [4 independent experiments](#).

Error bars represent the mean $\pm$ SEM. Means with different letters (a, b, c, d) or means with asterisk (*) show groups that are significantly different ($p < 0.05$) according to *F*-protected post hoc analysis.

Figure S1

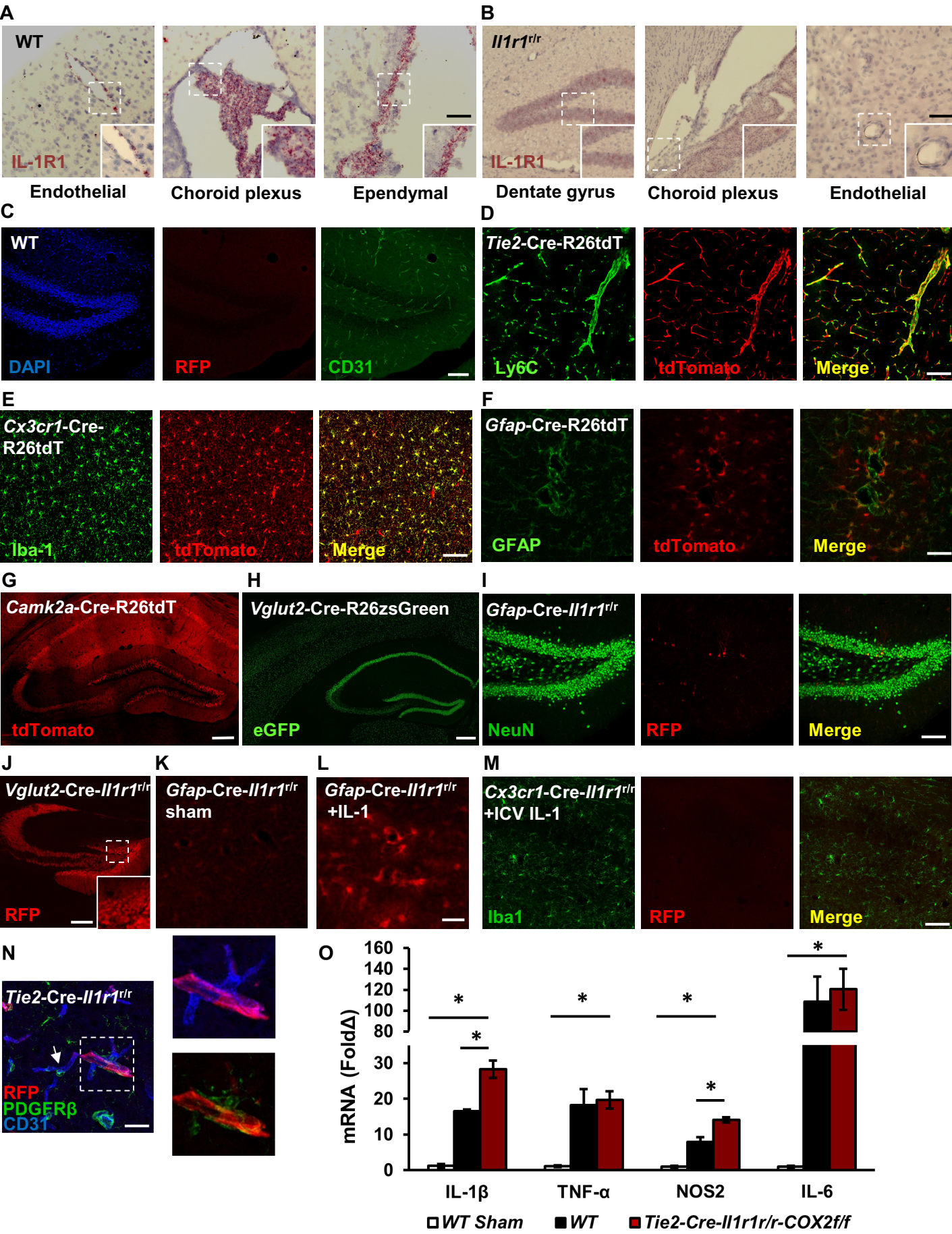


Figure S2

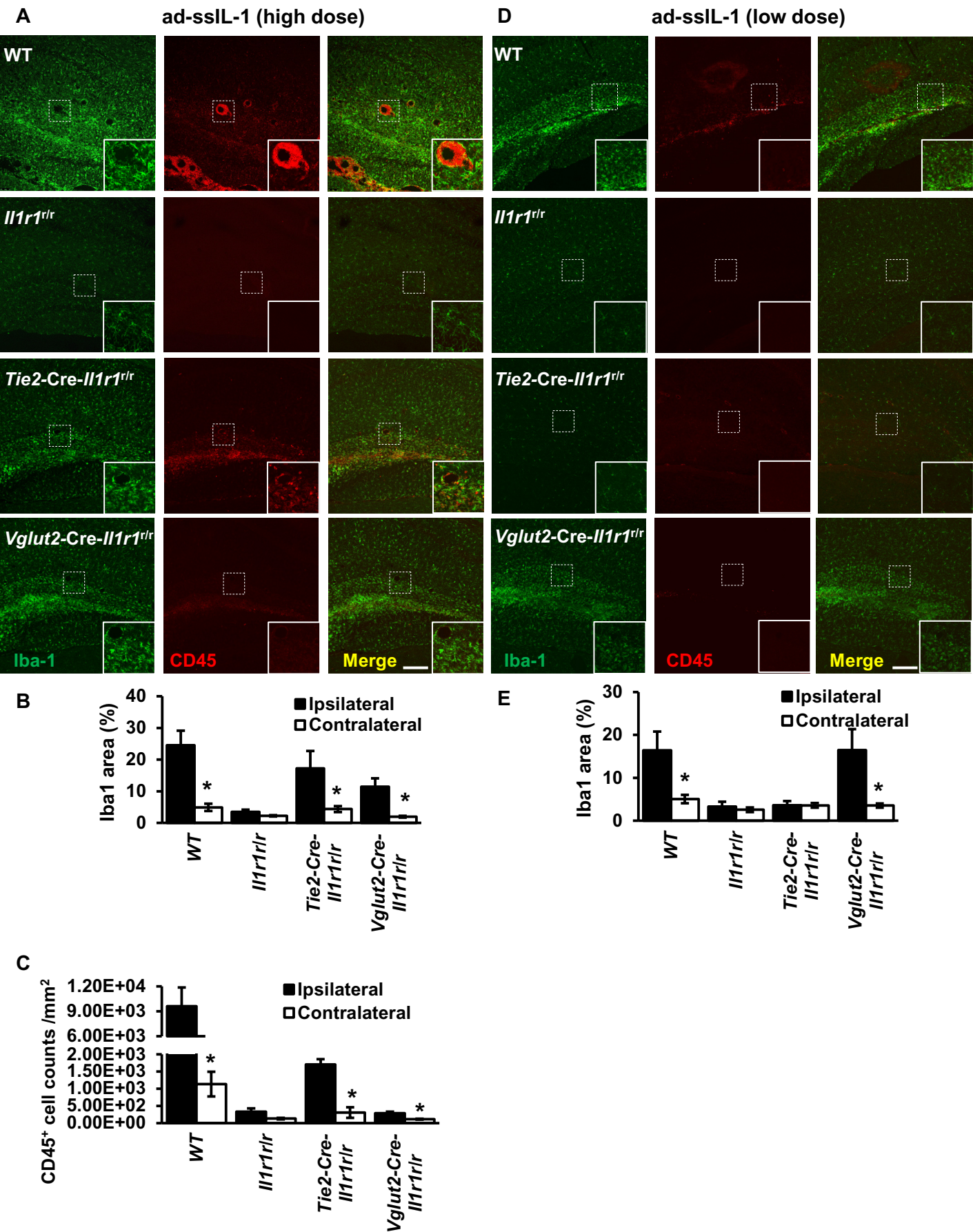


Figure S3

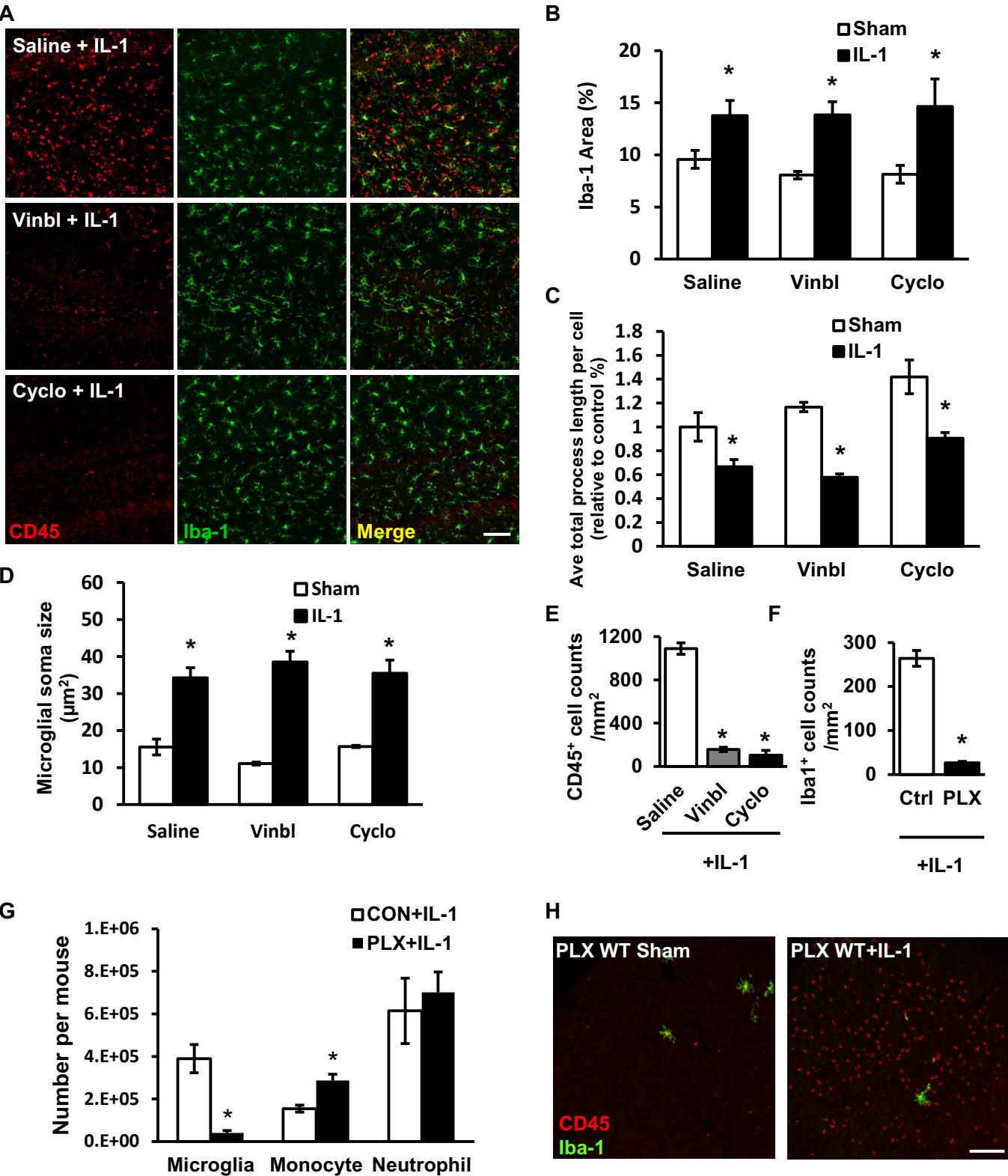


Figure S4

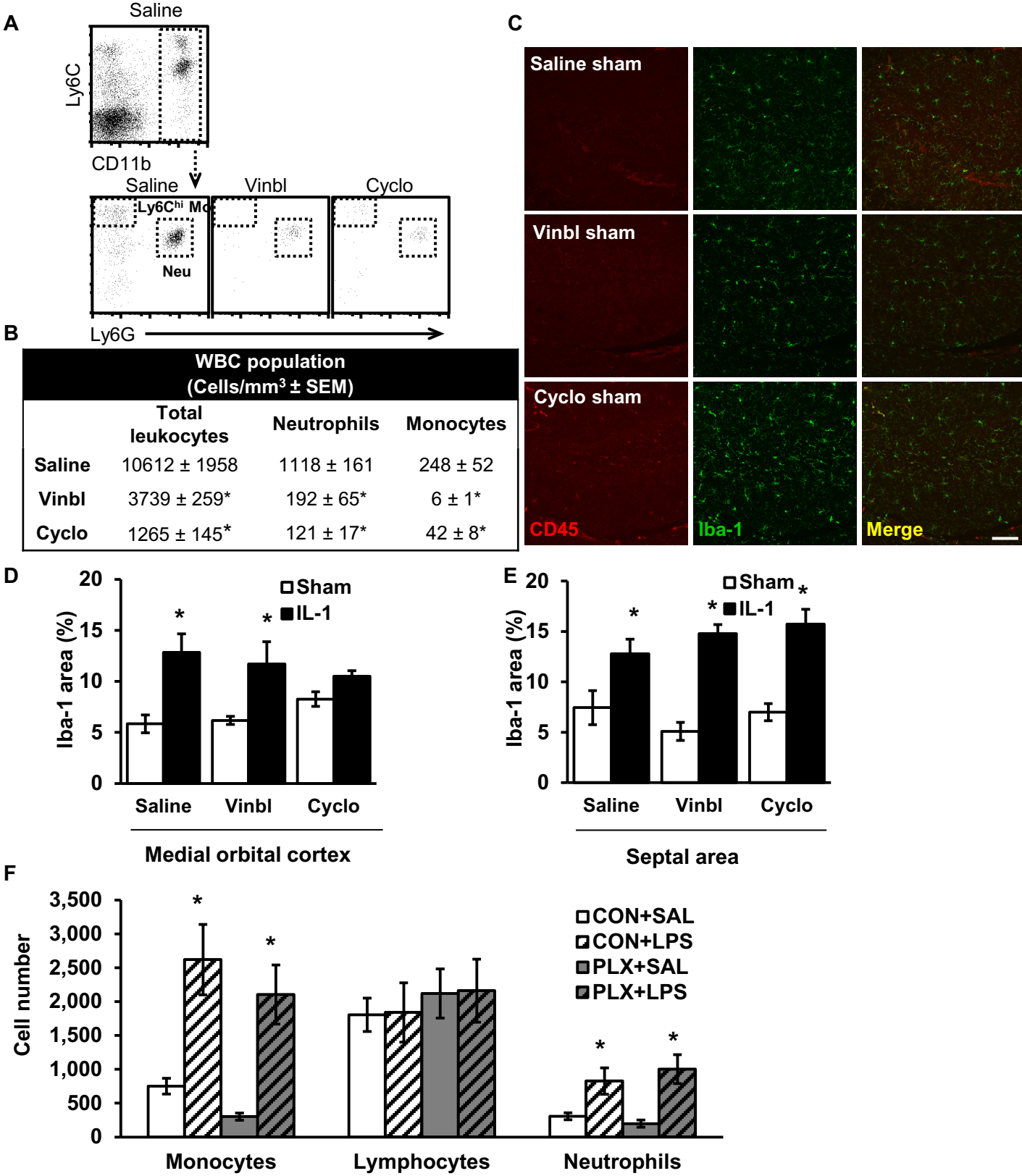


Figure S5

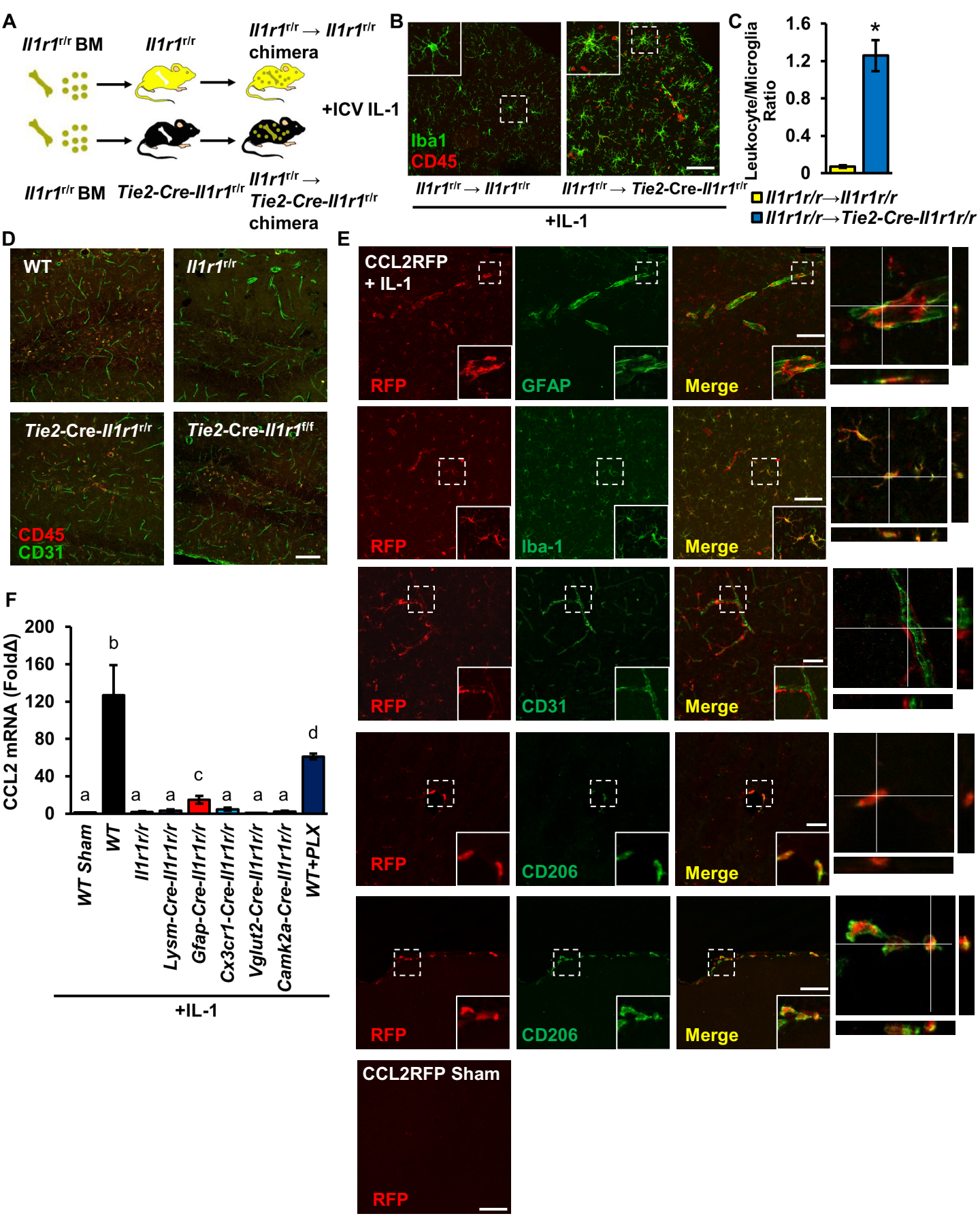


Figure S6

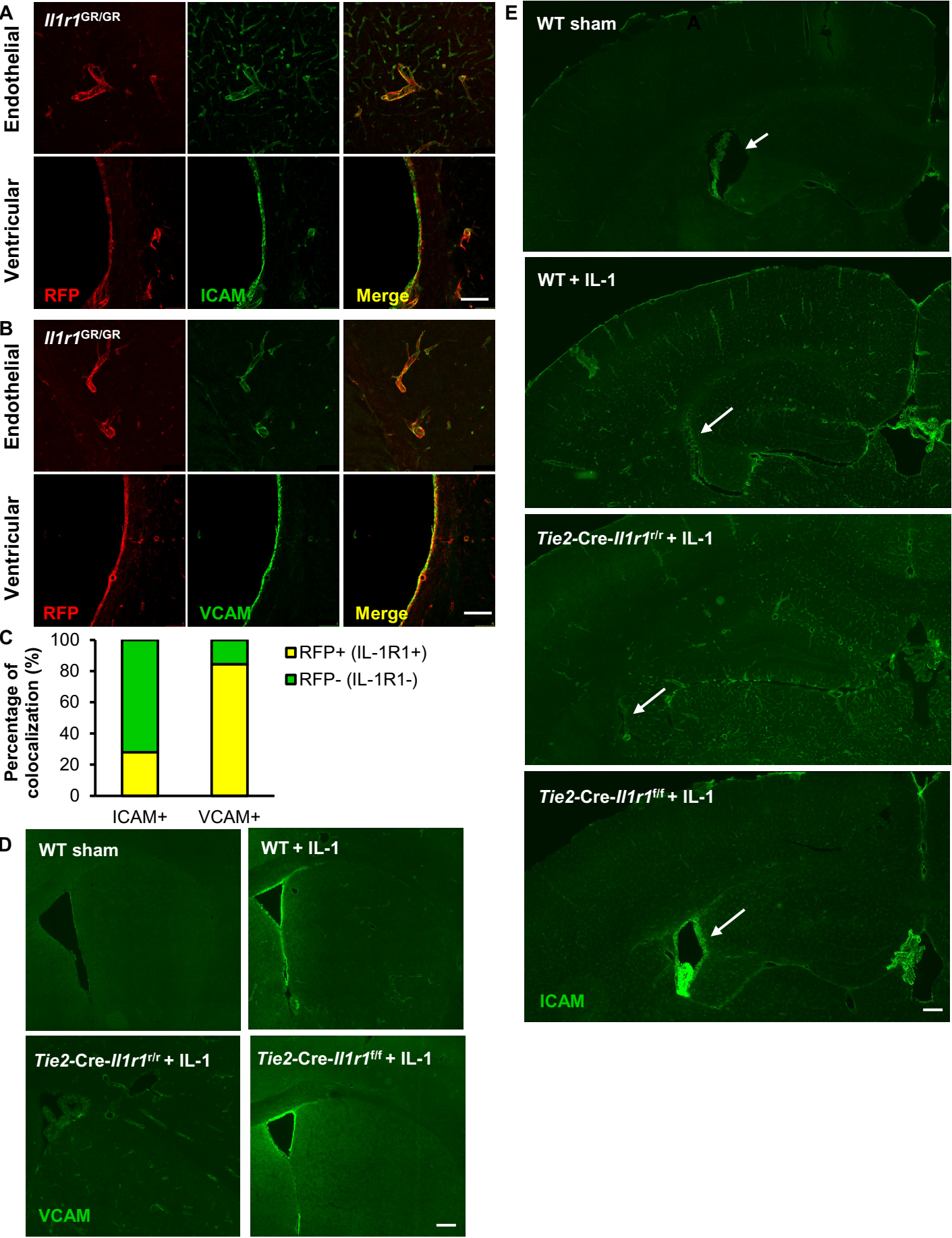


Figure S7

